Practical Significance

Why predictive machine learning matters in the real world

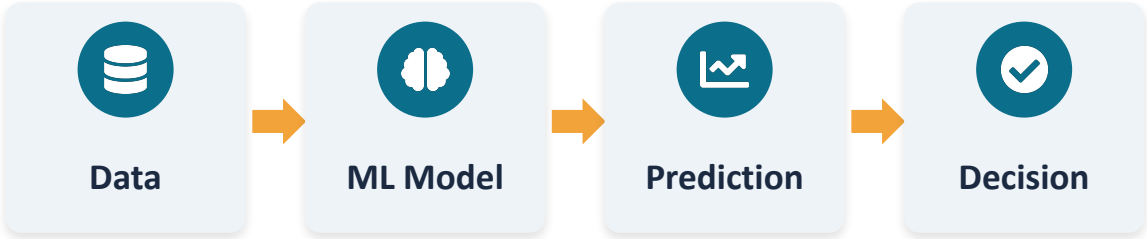
Early warning: raw data becomes timely predictions

Problems addressed: late diagnosis, slow disaster response, manual routine work

Better decisions: faster, evidence-based, consistent

Lower cost and error: automated screening and monitoring

This study: diabetes risk prediction as the case study



One approach, many applications

Disease Prediction

Early risk screening, e.g. diabetes

Disaster Management

Early warning and resource planning

Intelligent Automation

Predictive maintenance, smart decisions

Dataset

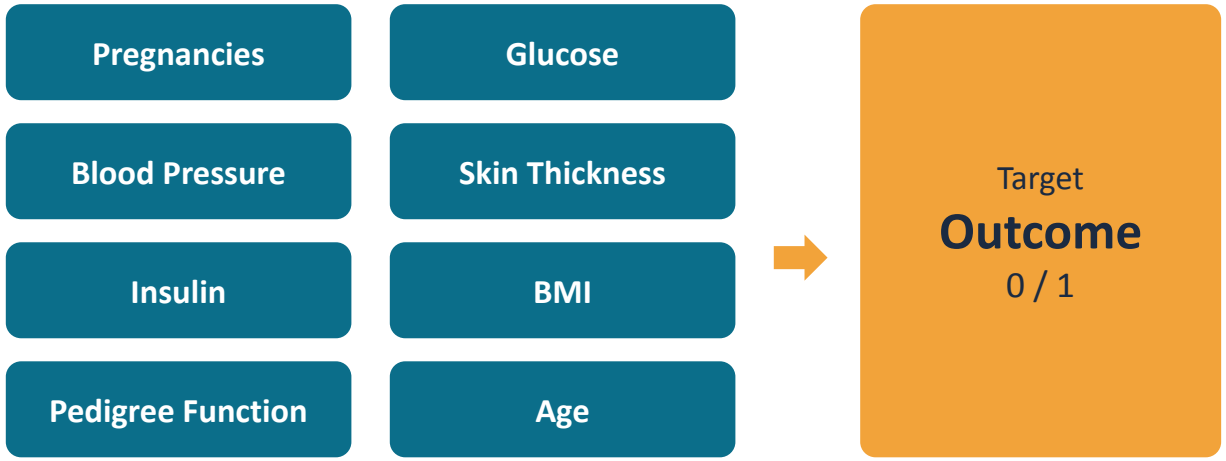
Case study: Pima Indians Diabetes dataset

768	8	Outcome	UCI
Records (patients)	Input features	Target: 0 = no, 1 = diabetes	Source: NIDDK, via UCI / Kaggle

Selection criteria

- Relevant to the research problem
- Diverse in values and cases
- Sufficiently large
- Reliable and representative

Dataset structure



Class balance: 500 negative vs 268 positive records

Literature Review

Purpose: understand what exists, so we know what is missing

Map existing work

Compare methods and results

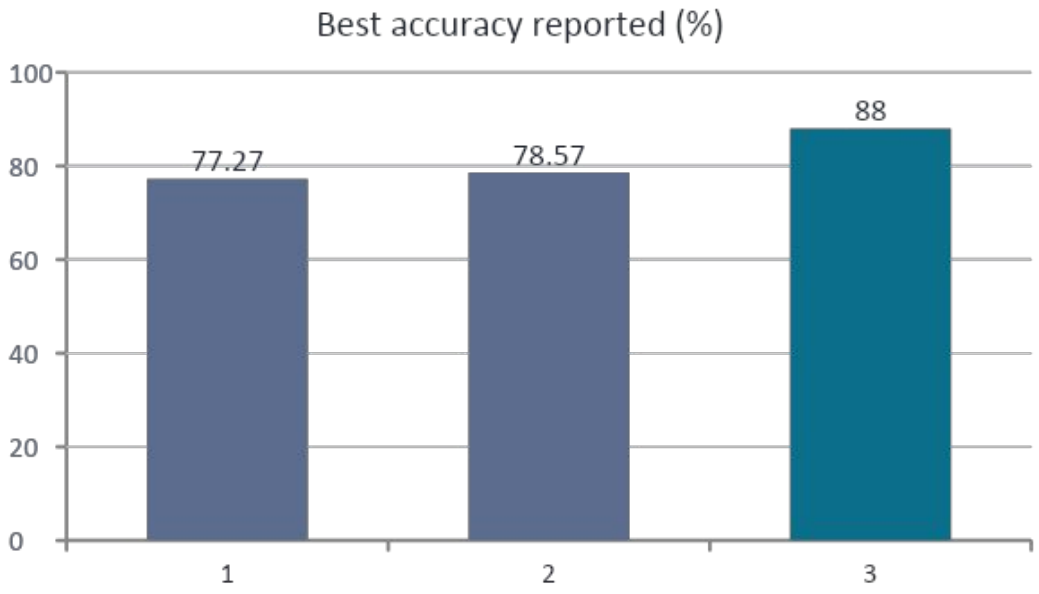
Identify the research gap

Study	Method / Models	Dataset	Key Finding
Alzboon et al. IEEE ComNet, 2023	Compared 8 ML models: LR, DT, RF, kNN, NB, SVM, GB, NN	Pima Indians, 768 patients	Neural network best (78.57%); RF next (76.30%)
Moussaoui, Salem, Mehaoua IEEE HealthCom, 2023	Several ML models with EDA and feature engineering; ROC/AUC, F1	Pima Indians diabetes data	SVM reached 77.27% accuracy
Early Diagnosis of Diabetes iJOE, 2023	Compared 8 ML models (LR, DT, RF, kNN, NB, SVM, GB, NN)	Pima Indians, 768 patients	LR and NN strongest overall; kNN and trees weaker
Paul et al. J. Basic Physical Research, 2025	LR, RF, SVM, XGBoost; accuracy, F1, AUC	Pima Indians	RF and XGBoost above 88% accuracy

Sources: published abstracts (IEEE Xplore, arXiv, iJOE, JBPR). Accuracy as reported by each study.

Comparative Analysis and Research Gap

Reviewed studies: what works, and what is still missing



Strengths

- Many models benchmarked
- Ensembles exceed 88% accuracy
- Standard metrics: F1, AUC

Limitations

- Single dataset (768 records)
- Results vary from 77% to 88%
- Generalisation not shown

Paul et al. report “above 88.0%”; shown as 88.

Gap: studies compare models on one small dataset with differing set-ups; consistent, validated, generalisable prediction is still missing.

Need: a standardised pipeline: cleaning, balancing, feature selection, ensemble model, cross-validation.

Proposed Contribution & Conclusion

Contribution: a validated, reproducible ML pipeline (case study: diabetes prediction)



Closes the gap

- Consistent, reproducible pipeline
- Handles missing values and class imbalance

Expected benefits

- More reliable, higher-recall prediction
- Earlier, low-cost screening support

Dataset fit

- Known benchmark: 768 records, 8 features
- One population: validate on more data

Key Takeaway

A validated ML pipeline turns routine data into early, reliable predictions, useful in healthcare, disaster management and intelligent automation.